

PROJECT PRESENTATION

AI-Powered IT/Admin Helpdesk Router

Intelligent ticket classification, RAG-based resolution, and smart escalation for faster, more reliable employee support.

TEAM

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Project Objectives

What the AI Helpdesk system is designed to achieve



Automate Helpdesk Management

Handle employee support requests efficiently end to end.



Intent Classification

Classify tickets by category — IT, HR, finance, or network.



Priority Assignment

Auto-assign urgency based on issue type and impact.



Duplicate Detection

Use semantic similarity to avoid repeated resolution work.



RAG-based Resolution

Retrieve knowledge base info and generate accurate answers.



Auto-Resolve Common Tickets

Resolve issues automatically when confidence is high.



Smart Escalation

Route complex/low-confidence tickets to Agent, L2, L3, or vendor.



Audit & Feedback Trail

Maintain ticket history, audit logs, and feedback for tracking.



Boost Support Efficiency

Reduce manual effort and improve response times overall.

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Problem Statement

- **Manual ticket handling is slow and error-prone.**

Organizations receive many IT and admin support requests daily, often in natural language, requiring manual effort to identify, categorize, and route each ticket.

- **This causes delays, misrouting, and overload.**

The process is time-consuming and increases workload on helpdesk staff, leading to delayed responses and incorrect ticket routing.

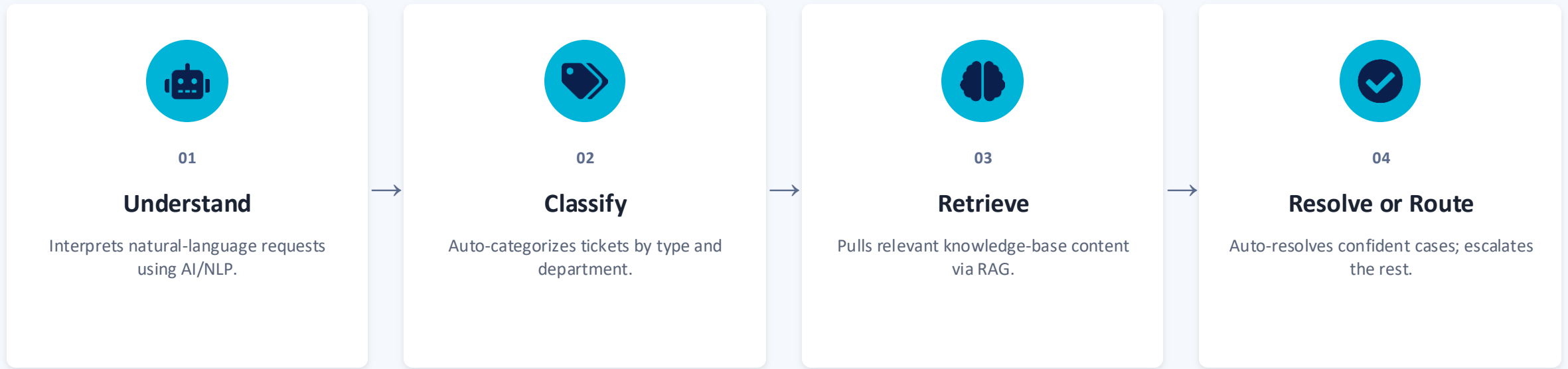
- **Known solutions go unused.**

Many common issues already have documented solutions in the knowledge base, yet employees still raise tickets — adding unnecessary work.

Proposed Solution

An Intelligent IT/Admin Helpdesk Router powered by AI

The system uses AI to understand user requests, automatically classify tickets, retrieve relevant solutions from a knowledge base using Retrieval-Augmented Generation (RAG), and deliver instant responses for common issues — routing anything it cannot resolve to the right support team.



Technology Stack

React.js, FastAPI, multi-agent AI, RAG, and SQLite

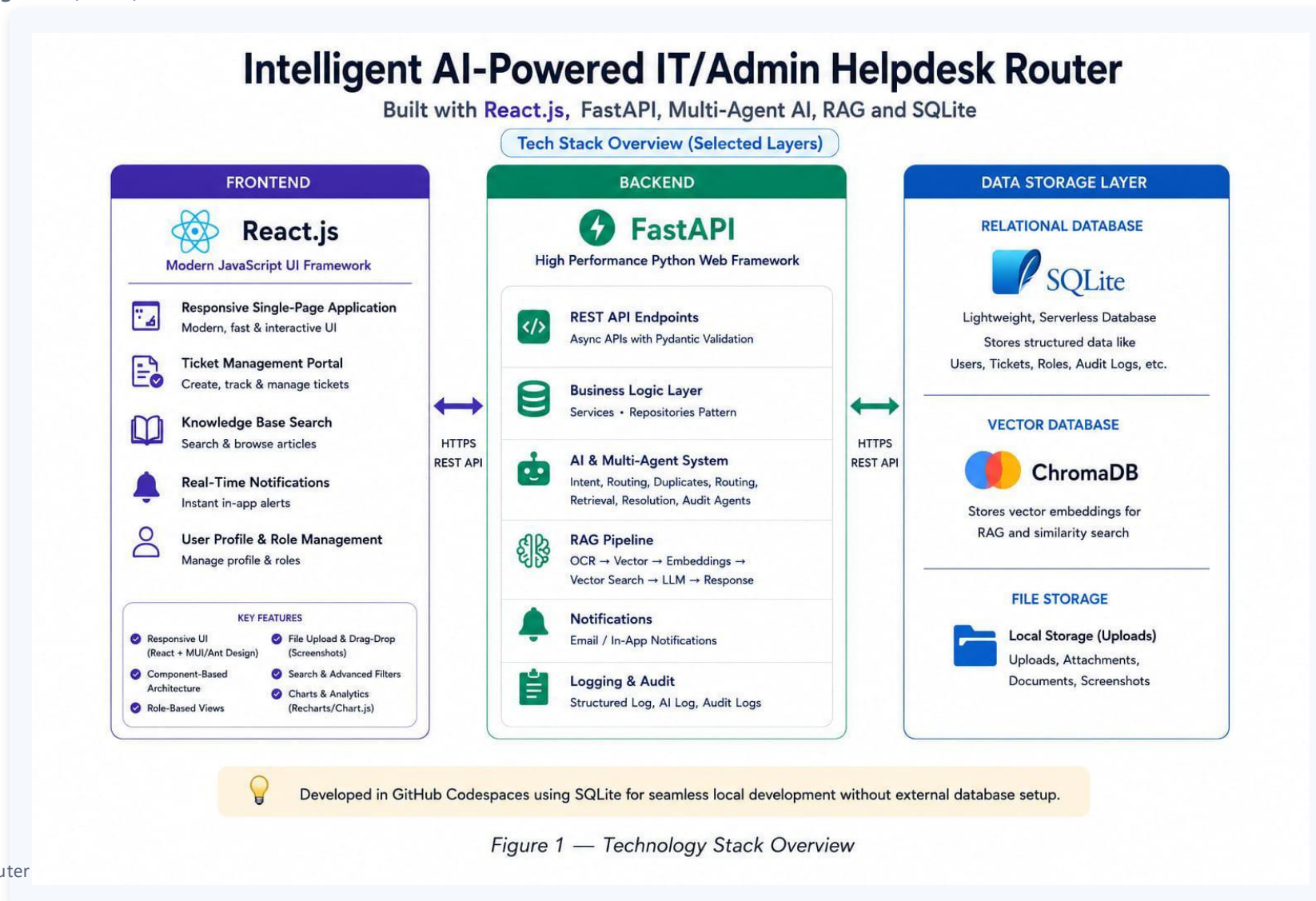
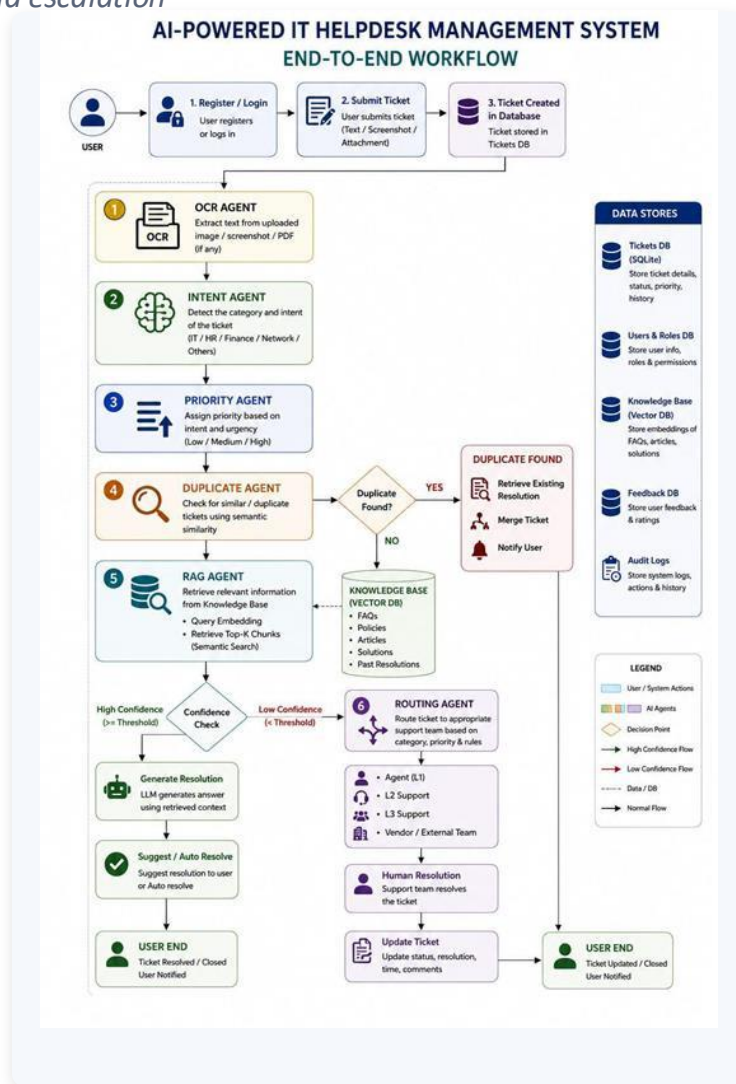


Figure 1 — Technology Stack Overview

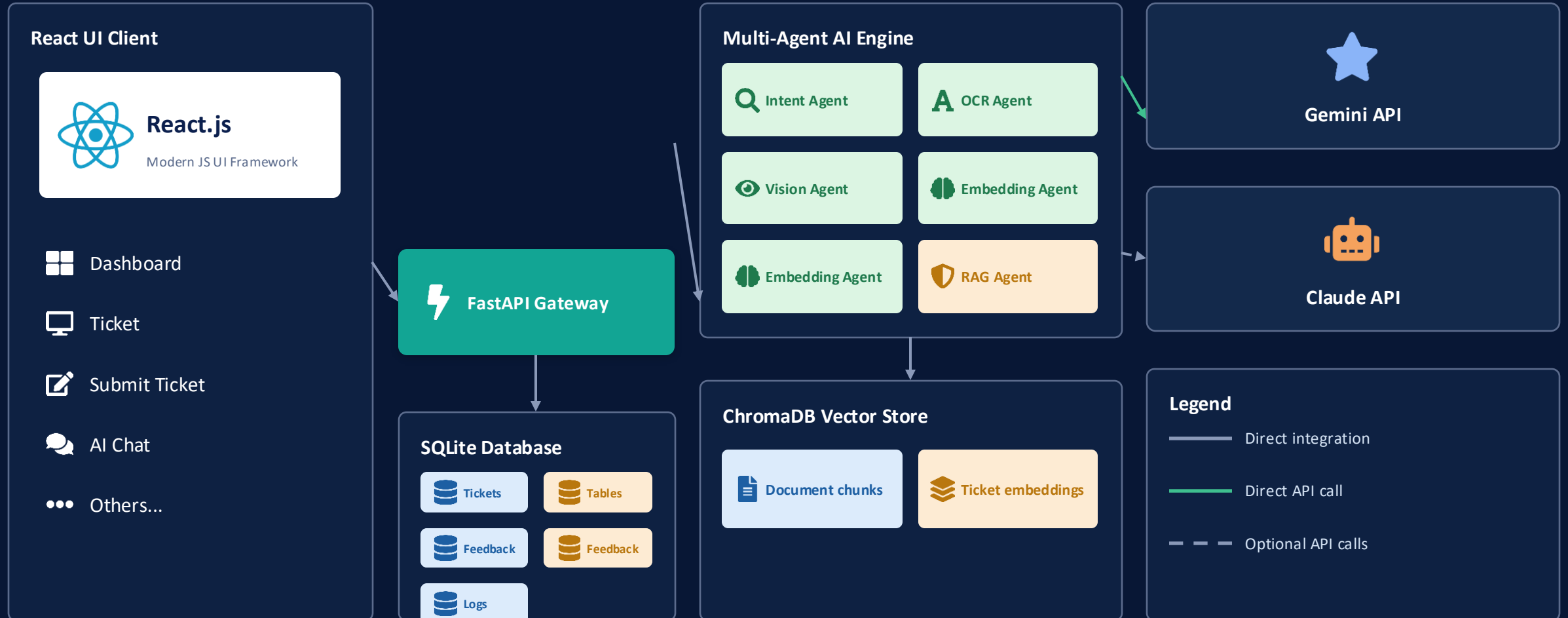
End-to-End Workflow

From ticket submission to resolution, duplicate handling, and escalation



System Architecture

React client, FastAPI gateway, multi-agent AI engine, and ChromaDB vector store

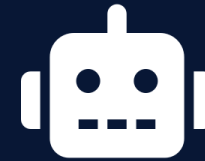


Conclusion

The AI Helpdesk Management System automates support ticket handling by combining classification, priority assignment, duplicate detection, and RAG-based resolution — reducing manual effort and routing complex tickets to the right team.

-  **Faster response time**
-  **Higher resolution accuracy**
-  **Full audit & feedback trail**
-  **Reliable escalation flow**

Thank You



Overall Impact

Improved response time, support efficiency, and ticket management accuracy — with reliable tracking end to end.